Simulation and Analysis of Robo-Matter: Active Brownian Particles with Chiral Dynamics and Magnetic Interactions

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by

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Declaration

I certify that the work contained in this report has been carried out by me under the guidance of my supervisor. The work has not been submitted to any other institute for any degree or diploma. I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

Whenever I have used material from other sources, including research papers, theoretical analysis, figures, computational ideas, and text, I have given due credit through citations and references. The thesis text has been written in my own words, and project-generated figures have been identified as simulation outputs. The simulations, interpretations, and analysis presented in this report are based on the computational work and material collected for this Bachelor Thesis Project.

Date: 1 May 2026 — Suraj
Place: Kharagpur — Roll No.: 22PH23011

Certificate

This is to certify that the project report entitled *Simulation and Analysis of Robo-Matter: Active Brownian Particles with Chiral Dynamics and Magnetic Interactions* submitted by Suraj (Roll No.: 22PH23011) to Indian Institute of Technology Kharagpur towards partial fulfilment of the requirements of the Bachelor Thesis Project is a record of bona fide work carried out under my supervision and guidance during Spring Semester 2026.

The work presented in this report involves the modelling, simulation, and analysis of Robo-Matter inspired by active magnetic Magbot systems. The student has studied the relevant literature, developed physics-based computational models, analyzed collective phase-like behavior, and documented the results in the present thesis report.

Date: 1 May 2026
Place: Kharagpur

Prof. Pintu Patra
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Acknowledgement

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Finally, I thank my peers and well-wishers for their discussions, feedback, and encouragement during the development and documentation of this work.

Abstract

Reconfigurable multifunctional materials are an important frontier in modern physics, materials science, and robotics because they seek to combine structural adaptability with active, programmable behavior. Recent experimental work on Robo-Matter has shown that simple robotic building blocks, called Magbots, can exhibit both matter-like properties such as self-assembly, active crystalline order, and phase-like transitions, and robot-like properties such as smart healing, morphing, force output, and infiltration. These behaviors arise because each building block possesses active motion, tunable magnetic interaction, and coupling to an external programmable light field.

This project develops a two-dimensional computational model of Robo-Matter using active magnetic disk particles inspired by Magbots. Each simulated Magbot is represented as a rigid circular body containing six magnetic sites around its perimeter. The particles possess position, body orientation, and internal magnet phase variables. Their dynamics combine self-propulsion, chiral rotation, light-field modulation, Gaussian light-gradient drift, soft steric repulsion, magnetic attraction, body torque, and internal magnet phase torque. A cell-list neighbor search is used to reduce unnecessary pairwise checks and make simulations of 100–170 Magbots computationally tractable.

The central physical question studied in this report is how collective organization emerges from the competition between activation, which agitates the system, and magnetic binding, which promotes local order. By varying activity, magnetic strength, interaction range, density, and noise, the simulation reproduces four representative phase-like regimes: active glass-like, active crystal, liquid-like, and gas-like states. The results are analyzed using local topological order, nearest-neighbor change rate, spatial snapshots, simulation videos, time-origin averaged mean squared displacement, and repeated random-seed MSD comparisons. This expanded evidence shows that strong magnetic binding with moderate activity supports ordered active-crystal-like behavior, strong binding with insufficient rearrangement produces arrested glass-like behavior, intermediate activity produces liquid-like restructuring with the largest displacement growth, and weak binding with high activity produces gas-like dispersion.

Overall, the work shows that a simplified physics-inspired model can capture several essential features of Robo-Matter: programmable activation, collective ordering, dynamic

bond rearrangement, self-organization, and phase-like transitions. The model is not an exact replica of the experimental system; rather, it is a compact computational platform for understanding the mechanisms by which simple active magnetic agents generate rich collective behavior. The project contributes a clear modelling framework, phase interpretation, parameter analysis, and simulation basis for future studies of adaptive robotic materials.

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1 Introduction

1.1 Research Context, Motivation, and Scope

Conventional materials are usually designed to perform a fixed function after fabrication. A crystalline solid may be strong, a polymer may be flexible, and a composite may be lightweight, but the internal building blocks of such materials generally do not make decisions, process information, or actively rearrange after deployment. This limitation has motivated the development of smart and active materials, where the material can respond to external stimuli such as light, temperature, mechanical loading, or chemical signals.

Active matter extends this idea further. It consists of many units that consume energy locally and convert it into motion or mechanical work. Examples include bacterial colonies, active colloids, vibrated granular particles, micro-swimmers, and robotic swarms. Unlike passive materials, active matter can organize itself far from equilibrium, generating clusters, vortices, waves, swirls, and coherent collective motion.

Robo-Matter is a recent concept that connects active matter with swarm robotics. In this framework, the material is composed of robotic building blocks that can move, interact, sense environmental signals, and reconfigure. The experimental Magbot system reported by Wang *et al.* demonstrates that small robotic units with magnetic interactions and light-field control can show both matter-like and robot-like behavior [1]. This dual character is the central inspiration for the present computational project.

Central idea

The project treats Robo-Matter as a many-body physics problem: if simple Magbot-like particles are given active motion, chirality, tunable magnetic binding, and light-field activation, then collective phases and material-like behavior can emerge without prescribing global order.

There is growing interest in materials that can adapt, heal, migrate, and change shape. Potential examples include protective surfaces that repair damage, robotic matter that navigates confined environments, microrobotic carriers for targeted transport, and modular systems that assemble into useful shapes on demand. Direct experimental construction of such systems can be expensive and technically challenging. A computational model is therefore valuable because it allows controlled variation of physical parameters and helps isolate the mechanisms responsible for observed behavior.

The motivation of this project is to understand how rich collective dynamics can arise from a minimal set of physically interpretable rules. Instead of modelling every hardware detail of a real Magbot, this work constructs a simplified active magnetic disk system. The simulation preserves the essential ingredients: circular bodies, multiple magnetic sites, chirality, light-dependent activity, torque, steric exclusion, and environmental control.

The broad problem addressed in this report is:

> *How can a physics-based simulation of active magnetic Magbot-like particles reproduce phase-like Robo-Matter behavior, and which model parameters control the transition between ordered, fluid, and dispersed collective states?*

This problem requires a model that is simple enough to simulate and interpret, but detailed enough to include both translational and rotational effects. It also requires analysis metrics that distinguish structural order from dynamic rearrangement.

The scope of the project is limited to two-dimensional simulations of rigid circular Magbot-like particles. Within this scope, the work develops a particle-level model with body motion, chirality, internal magnet phases, light-field coupling, magnetic interaction, and steric collision handling. The model is then used to simulate 100–150 interacting Magbots, identify active glass-like, active-crystal, liquid-like, and gas-like regimes, and interpret these regimes using local topological order and nearest-neighbor change rate. The comparison with experimental Robo-Matter is qualitative and mechanism-focused, because the present model is not a hardware-level replica of the full experimental platform.

The report first introduces the research area and literature, then states the rationale and objectives, presents the proposed model, analyzes the results, and closes with limitations and future directions. The appendices keep implementation details, parameter tables, and source-file information separate from the main narrative.

2 Literature Review

2.1 Relevant Literature and Research Gap

Smart materials respond to external stimuli by changing measurable properties such as shape, stiffness, optical response, or conductivity. Their importance lies in their ability to move beyond the static role of traditional materials. Active matter, by contrast, is defined by the continuous local consumption of energy by its constituents. This energy consumption drives the system away from thermal equilibrium and produces emergent collective states.

Active matter has been studied extensively in the context of colloids, bacteria, cell tissues, synthetic swimmers, and robotic particle systems. A common theme across these systems is that simple individual-level rules can generate large-scale order. Unlike equilibrium matter, where phases are determined by free-energy minimization, active phases are governed by an interplay among drive, interaction, noise, confinement, and dissipation.

Robo-Matter concept. Robo-Matter combines ideas from active matter, smart materials, and robot swarms. The experimental system of Wang *et al.* uses micro-robotic building blocks called Magbots [1]. These units are equipped with active rotation, magnetic binding sites, photosensitive control, and interaction with a programmable light field. The system is not simply a robot swarm, because it can also behave like a material with crystal-like, liquid-like, glass-like, and gas-like states. It is also not simply a passive material, because it can process environmental cues and actively reorganize.

The matter-like aspect of Robo-Matter appears in collective self-organization. Magbots can assemble into ordered structures, form active crystals, reorganize under changed activation, and display phase-like transitions. The same experimental work reports active glass-like, active-crystal, liquid-like, and gas-like states controlled primarily by the competition between activation strength and magnetic binding strength [1]. The robot-like aspect appears when Magbot assemblies respond to programmed light patterns. Reported demonstrations include active force output, smart healing, morphing, infiltration through barriers, cargo search, and robust operation under partial unit failure. These functions require information exchange between the building blocks and the environment; in the experiment this is achieved through camera tracking, an LED platform, and light-sensitive

Magbot response [1].

Active Brownian and chiral particle models. The Active Brownian Particle (ABP) model is a standard theoretical description of self-propelled particles. In its simplest form, each particle moves with a constant speed along an orientation vector that undergoes rotational diffusion. Chiral active particles include an additional intrinsic angular velocity, causing curved or circular motion. Such models are useful because they capture persistent motion, orientational decorrelation, ballistic-to-diffusive crossover, and collective effects under interaction.

The present project uses an ABP-inspired framework but extends it in three ways. First, the particles possess multiple magnetic sites instead of simple isotropic attraction. Second, each magnetic site has an internal phase variable that can rotate. Third, the particle activity is coupled to a moving Gaussian light field, mimicking programmable external activation.

Computational modelling of robotic swarms. Robotic-swarm simulations often emphasize communication, control laws, and task execution. Active-matter simulations emphasize physical interaction, noise, and emergent collective dynamics. Robo-Matter sits between these traditions. For this reason, the present simulation adopts a physics-first approach: it does not prescribe a desired global structure, but allows local forces and torques to produce organization.

Why simulation matters

Experiments demonstrate what Robo-Matter can do. Simulation helps answer why it happens: which terms in the equations promote binding, which terms melt order, how light-field modulation acts like activation, and how torque couples local magnet alignment to whole-body rotation.

The experimental literature demonstrates Robo-Matter using physical Magbots and a sophisticated feedback platform. However, a beginner-friendly computational model that connects the experimental idea to explicit equations of motion is useful for learning and further exploration. The gap addressed here is the development and interpretation of a compact, physics-based simulation that reproduces key qualitative phase regimes while remaining transparent enough to modify.

3 Rationale of the Study

3.1 Rationale and Novelty

Robo-Matter provides a route to materials that can change their structure and function after deployment. Understanding such systems requires a blend of mechanics, statistical physics, nonlinear dynamics, and robotics. A computational study is especially appropriate because the phase behavior depends on many coupled parameters: activity, attraction, density, noise, chirality, confinement, and light control.

This project matters because it transforms the Robo-Matter idea from a qualitative concept into an explicit simulation model. The model allows one to ask: what happens if magnetic attraction is increased? What if the light field is stronger? What if the box is made larger? What if rotation becomes faster? These questions are difficult to answer from static figures alone but can be explored through parameter sweeps.

The novelty of this BTP work is not the invention of Robo-Matter itself, which is credited to the recent experimental literature [1]. The contribution is a compact computational realization in which each Magbot-like body has six magnetic sites, active propulsion, chirality, light-dependent activation, internal magnet phase dynamics, and torque coupling. The simulation is deliberately transparent: every important physical contribution appears as a term in the update equations, while the cell-list method keeps neighbor detection efficient enough for 100–150 particles. This makes the model useful for testing how separate mechanisms such as attraction, noise, density, and light modulation influence the final collective state.

The central competition in the model is between activation and magnetic binding:

$$\text{collective state} \sim \frac{\text{magnetic binding and ordering}}{\text{active agitation and noise}}. \tag{3.1}$$

When binding dominates, particles stick and order. When activation dominates, bonds break and the system becomes fluid or gas-like. Between these extremes, the system can produce nontrivial dynamic structures.

Interpretive principle

The same interaction rules can generate different material states. The observed phase is not hard-coded; it emerges from the relative strength of activity, attraction, density, and noise.

4 Objectives

The objective of this project is to build and interpret a physics-based simulation of Robo-Matter inspired by Magbot systems. The study begins from the experimental idea of active magnetic robotic building blocks [1], translates the relevant mechanisms into a two-dimensional active-particle model, and then uses the model to study phase-like organization.

Table 4.1: Research objectives of the BTP work.

Objective area	Implementation in this project
Model construction	A two-dimensional Magbot-like particle model is built with active translation, chiral rotation, light-dependent activation, six magnetic sites, internal magnet phases, magnetic torque, and steric repulsion.
Computational method	A cell-list neighbor search is used so that magnetic and contact interactions are evaluated locally rather than through unnecessary all-pair checks.
Phase study	Representative active glass-like, active-crystal, liquid-like, and gas-like regimes are generated by changing activity, attraction, density, and interaction range.
Analysis	Local topological order and nearest-neighbor change rate are used to distinguish structural ordering from dynamic rearrangement.
Literature connection	The simulated mechanisms and phase terminology are compared qualitatively with the Robo-Matter experiments of Wang *et al.* [1].

5 Materials and Methods

5.1 Model Formulation

The simulation represents Robo-Matter as a collection of rigid circular particles in a square two-dimensional domain. Each particle is a Magbot-like unit with a body center, body angle, and six magnetic sites arranged around its perimeter. The system is evolved using an overdamped discrete-time update, appropriate for a strongly dissipative robotic or granular environment where inertia is not the dominant modelling feature.

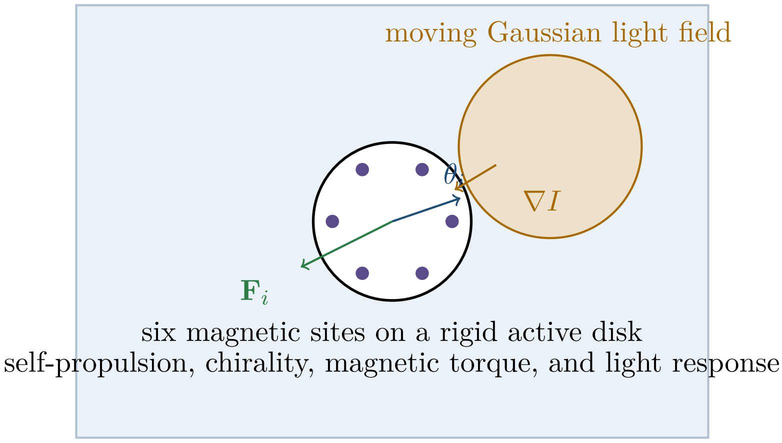

Figure 5.1: Schematic of the simulated Magbot model. The drawing was prepared for this report; the six-site active magnetic body and light-field control idea are conceptually inspired by the Robo-Matter Magbot system reported in Ref. [1].

State variables. For the ith Magbot, the main state variables are:

$$\mathbf{x}_i(t) = (x_i(t), y_i(t)), \qquad \theta_i(t), \qquad \phi_{ik}(t), \quad k = 1, \ldots, N_{\text{mag}}. \tag{5.1}$$

Here $\mathbf{x}_i$ is the body-center position, θ_i is the body orientation, and ϕ_{ik} is the internal phase of the kth magnetic site. The fixed angular location of the kth site on the body is denoted by α_k. The absolute magnet angle is therefore

$$\psi_{ik} = \theta_i + \alpha_k + \phi_{ik}. \tag{5.2}$$

The six fixed anchor points are placed at radius R_{ring} from the center:

$$\mathbf{a}_k = R_{\mathrm{ring}} \begin{bmatrix} \cos\alpha_k \\ \sin\alpha_k \end{bmatrix}, \qquad \alpha_k = \frac{2\pi(k-1)}{N_{\mathrm{mag}}}. \tag{5.3}$$

Light field. The simulation uses a Gaussian light field to represent a programmable activation pattern. For a light center $\mathbf{x}_L(t)$, the intensity at position $\mathbf{x}$ is

$$I(\mathbf{x}, t) = I_0 + A_L \exp\left[-\frac{|\mathbf{x} - \mathbf{x}_L(t)|^2}{2\sigma_L^2}\right]. \tag{5.4}$$

For the moving-spot case,

$$\mathbf{x}_L(t) = R_L \begin{bmatrix} \cos(2\pi t/T_L) \\ \sin(2\pi t/T_L) \end{bmatrix}. \tag{5.5}$$

The intensity is clipped to the interval $[0, 1]$ in the implementation. The light affects both translational and rotational motion. It increases the active speed and changes the chiral angular velocity:

$$v_i = v_{\mathrm{base}} + \gamma I_i, \tag{5.6}$$

$$\omega_i = \chi_i \left(\omega_{\mathrm{base}} + \beta I_i\right), \tag{5.7}$$

where $\chi_i = -1$ denotes clockwise rotation and $\chi_i = +1$ denotes counterclockwise rotation.

Light-gradient drift. The model includes a simplified drift toward brighter regions:

$$\mathbf{F}_{L,i} = k_L \nabla I(\mathbf{x}_i, t). \tag{5.8}$$

For the Gaussian component,

$$\nabla I = I_{\mathrm{spot}} \frac{\mathbf{x}_L - \mathbf{x}}{\sigma_L^2}. \tag{5.9}$$

This rule is not intended as a full electromechanical model of the real photoresistor-motor coupling. It is a compact way to represent light-guided activation and migration.

Magnetic interaction. Each Magbot has six magnetic sites. For a pair of magnetic sites separated by vector $\mathbf{r}$ and distance r, the simulation uses the simplified interaction energy

$$U_{ab} = -k_m \exp\left(-\frac{r^2}{r_0^2}\right)\cos(\Delta\psi), \tag{5.10}$$

where k_m is magnetic strength, r_0 is the decay scale, and $\Delta\psi$ is the relative magnet-angle difference. The interaction is evaluated only when $r < r_{\text{cut}}$.

The force contribution is obtained from the gradient of this potential. In the implemented form,

$$\mathbf{F}_{ab} = \frac{2k_m}{r_0^2}\exp\left(-\frac{r^2}{r_0^2}\right)\cos(\Delta\psi)\,\mathbf{r}. \tag{5.11}$$

The internal magnet phase torque is

$$\tau_\phi = k_m \exp\left(-\frac{r^2}{r_0^2}\right)\sin(\Delta\psi). \tag{5.12}$$

When a magnetic force acts away from the body center, it also creates body torque:

$$\tau_{\text{body}} = \mathbf{r}_{\text{lever}} \times \mathbf{F}. \tag{5.13}$$

Steric repulsion and collision handling. To prevent unphysical overlap, a hard-core repulsive force is applied when two disks overlap. If d_{ij} is the center-to-center distance,

$$\mathbf{F}_{\text{rep}} = -k_{\text{rep}}(2R_{\text{bot}} - d_{ij})\hat{\mathbf{r}}_{ij}, \qquad d_{ij} < 2R_{\text{bot}}. \tag{5.14}$$

After force-based integration, the code also performs direct overlap resolution through a few sweeps. This improves numerical stability in dense configurations.

Equations of motion. The overdamped translational update is

$$\mathbf{x}_i(t+\Delta t) = \mathbf{x}_i(t) + \left[\mu_t \mathbf{F}_i + \mathbf{F}_{L,i} + v_i \mathbf{n}_i\right]\Delta t + \sqrt{2D_t\Delta t}\,\boldsymbol{\eta}_i, \tag{5.15}$$

where $\mathbf{n}_i = (\cos\theta_i, \sin\theta_i)$ and $\boldsymbol{\eta}_i$ is a Gaussian random vector.

The body angle update is

$$\theta_i(t+\Delta t) = \theta_i(t) + \left[\omega_i + \mu_r \tau_i\right]\Delta t + \sqrt{2D_r\Delta t}\,\xi_i. \tag{5.16}$$

The internal magnet phase update is

$$\phi_{ik}(t+\Delta t) = \phi_{ik}(t) + \mu_\phi \tau_{\phi,ik}\Delta t + \sqrt{2D_\phi\Delta t}\,\zeta_{ik}. \tag{5.17}$$

5.2 Numerical Implementation and Analysis

The main simulations use reflecting square boundaries. If a particle crosses a vertical wall, its orientation is reflected as

$$\theta \to \pi - \theta. \tag{5.18}$$

If it crosses a horizontal wall,

$$\theta \to -\theta. \tag{5.19}$$

The position is clipped inside the simulation box. This mimics elastic reflection at rigid boundaries.

Cell-list neighbor search. Direct pairwise interaction checking scales as $O(N^2)$ and becomes inefficient for large N. The simulation therefore divides the domain into cells whose side length is comparable to the interaction cutoff. Each particle checks only particles in its own cell and neighboring cells. This keeps the interaction search efficient while preserving local pair detection.

Computational design choice

The cell-list method is important because the model contains both body-level and site-level interactions. Without local neighbor filtering, checking all possible site pairs for all Magbot pairs would become unnecessarily expensive.

Analysis metrics. The results are analyzed using two complementary structural metrics, whose mathematical definitions are given in Chapter 6. The first metric is a local sixfold order parameter, denoted r_6, which measures whether the neighbor directions around each Magbot resemble a locally ordered sixfold packing. The second metric is the nearest-neighbor change rate, denoted n_c, which measures how rapidly the neighbor list of each Magbot changes with time. Together these quantities separate static-looking order from genuine dynamical stability.

Local topological order. The local topological order parameter measures the degree of local structural organization. A high value indicates that the neighbors of a particle are arranged in a preferred sixfold geometry, which is the natural local symmetry for circular particles and for the six-site Magbot design. In the plots generated for this project, a threshold line near 0.30 is used as a practical visual separator between low-order gas-like or fluid-like states and more ordered regimes.

Nearest-neighbor change rate. The nearest-neighbor change rate n_c measures how frequently the local neighborhood of particles changes. A low value indicates persistent contacts, which may occur in a crystal-like or arrested glass-like state, while a high value indicates frequent bond breaking and rearrangement. In the plots, a threshold line near $n_c = 5$ is used as a practical separator between relatively stable and actively reorganizing structures.

Mean squared displacement and repeated measurements. Mean squared displacement is used as a mobility metric. For a trajectory $\mathbf{x}_i(t)$, the time-origin averaged MSD is computed as

$$\mathrm{MSD}(\tau) = \left\langle |\mathbf{x}_i(t+\tau) - \mathbf{x}_i(t)|^2 \right\rangle_{i,t}, \tag{5.20}$$

where the average is taken over particles and all valid starting times. This means that each MSD curve contains many repeated displacement measurements from the same phase trajectory. In addition, three independent random-seed simulations were run for each phase to compare the repeated-seed MSD trend and estimate the variability of the mobility result.

5.3 Simulation Cases

Representative parameter choices used in the project are summarized in Table 5.1. These values are extracted from the simulation scripts in the `states` folder.

Table 5.1: Representative parameter sets for four simulated Robo-Matter regimes.

Regime	N	box	v_{base}	ω_{base}	k_m	r_0
Active glass-like	120	17.0	0.70	0.00	10.00	1.50
Active crystal	170	17.0	10.45	6.90	10.00	5.00
Liquid-like	100	20.0	8.00	7.00	1.15	1.40
Gas-like	100	20.0	10.00	10.00	0.80	0.80

6 Theory

6.1 Active Brownian and Chiral Motion

A minimal active Brownian particle has the dynamics commonly used to model self-propelled units in soft active matter [2, 3]:

$$\dot{\mathbf{x}} = v_0\mathbf{n}(t) + \sqrt{2D_t}\,\boldsymbol{\eta}(t), \tag{6.1}$$

$$\dot{\theta} = \sqrt{2D_r}\,\xi(t). \tag{6.2}$$

At short times, the particle keeps its direction and motion is approximately ballistic. At long times, rotational diffusion destroys directional memory and the motion becomes diffusive. This crossover is central to active matter.

Chiral extension. For a chiral particle, an intrinsic angular velocity is added:

$$\dot{\theta} = \omega + \sqrt{2D_r}\,\xi(t). \tag{6.3}$$

If noise and interaction are absent, the particle follows a circular trajectory with approximate radius

$$R_{\text{circle}} = \frac{v_0}{|\omega|}. \tag{6.4}$$

In the Robo-Matter simulation, chirality is important because Magbots in the experimental system rotate due to vibration motors and tilted brushes. The sign of ω controls clockwise or counterclockwise motion.

Mean squared displacement. Although the primary project plots focus on structural order and neighbor-change rate, the theoretical mean squared displacement (MSD) remains a useful reference quantity. For a non-chiral ABP in two dimensions,

$$\langle \Delta r^2(t) \rangle = 4D_t t + \frac{2v_0^2}{D_r^2}\left(D_r t + e^{-D_r t} - 1\right). \tag{6.5}$$

For $t \ll D_r^{-1}$, the leading active contribution is approximately $v_0^2 t^2$, indicating ballistic motion. For $t \gg D_r^{-1}$, the MSD becomes diffusive with an effective diffusion coefficient

$$D_{\text{eff}} = D_t + \frac{v_0^2}{2D_r}. \tag{6.6}$$

In interacting Robo-Matter, this simple expression is modified by magnetic binding, confinement, collision, and collective rearrangement. However, the ballistic-to-diffusive idea remains useful for interpreting activity-driven spreading.

6.2 Structural Order and Neighbor-Exchange Theory

The phase classification in this project cannot be based only on visual snapshots. A final image can show clusters, but it cannot reveal whether those clusters are stable, flowing, or kinetically trapped. For this reason, two local diagnostics are used together with MSD: a sixfold local order parameter and a nearest-neighbor change parameter. The sixfold order metric is analogous to the bond-orientational order parameters commonly used for two-dimensional particle assemblies and active-particle systems [3]. In the present simulation it is adapted to the Magbot geometry, where each body is circular and carries six magnetic sites.

At a measurement time t_m, the neighbor set of Magbot i is defined as

$$\mathcal{N}_i(t_m) = \{ j \neq i : |\mathbf{x}_j(t_m) - \mathbf{x}_i(t_m)| \leq r_{\text{nn}} \}, \tag{6.7}$$

where r_{nn} is the analysis cutoff. In the project-generated parameter plots, $r_{\text{nn}} = 2.35$ is used. This cutoff is slightly larger than the hard body contact diameter, so it counts particles that are close enough to behave as local structural neighbors without counting distant particles that are not part of the same local arrangement.

For every neighbor $j \in \mathcal{N}_i(t_m)$, the bond angle from particle i to particle j is

$$\alpha_{ij}(t_m) = \operatorname{atan2}\left(y_j(t_m) - y_i(t_m), x_j(t_m) - x_i(t_m)\right), \tag{6.8}$$

where the two-argument arctangent preserves the correct quadrant of the bond direction. The complex sixfold local order around particle i is then

$$\psi_6^{(i)}(t_m) = \frac{1}{N_i(t_m)} \sum_{j \in \mathcal{N}_i(t_m)} \exp[6\,\mathrm{i}\,\alpha_{ij}(t_m)]\,, \qquad N_i(t_m) = |\mathcal{N}_i(t_m)|. \tag{6.9}$$

If a particle has fewer than two neighbors, the simulation assigns $\psi_6^{(i)} = 0$ because an isolated particle or a single pair cannot define a meaningful local angular structure. The factor of six makes the metric insensitive to rotations by 60°, which is appropriate for locally hexagonal packing and for six-site Magbot symmetry. The magnitude $|\psi_6^{(i)}|$ lies between zero and one. It approaches one when the local bond directions around particle i are close to sixfold angular order, and it decreases when the neighbor directions are irregular.

The scalar order parameter plotted in the report is the particle average

$$r_6(t_m) = \frac{1}{N} \sum_{i=1}^{N} \left|\psi_6^{(i)}(t_m)\right|. \tag{6.10}$$

This definition deliberately averages the magnitude of the local order rather than the complex value itself. Therefore, different ordered domains with different absolute orientations do not cancel each other. This is important for Robo-Matter because the active glass-like and active-crystal states may contain locally ordered clusters whose global orientations are not identical. The metric therefore measures local structural organization, not perfect global crystallinity.

The nearest-neighbor change parameter measures the time dependence of the same neighbor sets. Between two consecutive measurement times t_{m-1} and t_m, the number of neighbor-list changes for particle i is computed using the symmetric difference

$$C_i(t_m) = |\mathcal{N}_i(t_m) \triangle \mathcal{N}_i(t_{m-1})|\,, \tag{6.11}$$

where $\triangle$ denotes the set of particles that are present in one neighbor set but not in the other. This counts both newly formed neighbors and lost neighbors. The project reports the average rate

$$n_c(t_m) = \frac{1}{N\,\Delta t_m} \sum_{i=1}^{N} C_i(t_m), \qquad \Delta t_m = t_m - t_{m-1}. \tag{6.12}$$

In the code, the metric is evaluated every five simulation steps, so $\Delta t_m = 5\Delta t$. The units of n_c are neighbor-list changes per particle per unit simulation time. A large value means that particles are frequently exchanging local contacts, which is expected in liquid-like and gas-like states. A small value means that the local contact network is persistent, which can indicate an ordered crystal or an arrested glass. The distinction between these two cases comes from reading n_c together with r_6 rather than using either metric alone.

Why both parameters are required

A high value of r_6 alone does not prove that the system is a healthy active crystal, because a glass-like cluster can also have local order. Similarly, a high value of n_c alone does not prove useful fluidity, because a gas-like state can have frequent contact changes without stable local structure. The physically meaningful classification comes from the pair (r_6, n_c): active crystals show high order with neighbor exchange mainly during early annealing, active glasses show local order with suppressed rearrangement, liquids show persistent exchange with partial order, and gases show low order with mobile disorder.

6.3 Phase Interpretation

The phase behavior can be interpreted through a dimensionless qualitative ratio:

$$\Lambda = \frac{k_m}{v_{\text{base}} + \omega_{\text{base}} + \epsilon}, \tag{6.13}$$

where ϵ prevents division by zero and reminds us that this is not a rigorous thermodynamic parameter. Large Λ indicates strong binding compared with agitation. Small Λ indicates weak binding compared with activity. In the project, this ratio is used conceptually rather than as a fitted universal order parameter.

Table 6.1: Qualitative transition logic used to interpret simulated phases.

Condition	Expected behavior	Representative phase
Strong binding, low activity	Particles bind but cannot easily escape disordered local traps.	Active glass-like
Strong binding, moderate activity	Motion anneals defects while attraction maintains local order.	Active crystal
Comparable binding and activity	Bonds form and break continually; the network flows.	Liquid-like
Weak binding, high activity	Particles or small clusters roam independently.	Gas-like

Role of magnet symmetry. Six magnetic sites produce a natural route to local hexagonal or honeycomb-like packing. The fixed body anchors introduce anisotropy: attraction depends not only on distance but also on relative magnet orientation. This makes the model richer than a simple isotropic Lennard-Jones-like attraction. Body torque and internal magnet torque allow local magnetic alignment to influence both individual particle orientation and collective structure.

Relation to experimental Robo-Matter. The experimental paper reports that homogeneous light intensity can play a temperature-like role: stronger light increases Magbot activation and changes collective organization [1]. In this simulation, the light field increases active speed and rotation rate. Therefore, activity can melt stable clusters, promote reorganization, or disperse the system depending on magnetic binding strength.

7 Results and Discussion

7.1 Expanded Evidence Set

The results are presented as a phase-by-phase analysis rather than as a short collection of final plots. Each phase is supported by four forms of evidence: a parameter plot showing local topological order and nearest-neighbor change rate, a spatial simulation snapshot showing the arrangement of Magbots, an MSD plot measuring mobility, and a linked simulation video showing the full time evolution. This is important because a single final frame can be misleading. A structure that looks ordered may still be dynamically rearranging, and a dispersed state may still have short-lived local contacts. The combined evidence therefore separates structure, motion, and time evolution.

The phase labels used in this chapter follow the qualitative Robo-Matter terminology of Wang *et al.* [1]. The present figures are project-generated simulation outputs and should be read as mechanism-level results, not as direct experimental calibration. The main purpose is to determine how magnetic binding, activity, density, and chirality combine to produce different nonequilibrium material states.

Table 7.1: Evidence included for each simulated Robo-Matter phase. The video links point to the BTP-2 GitHub folder.

Regime	Plots and snapshots used in the report	Simulation video
Active crystal	Parameter trace, spatial snapshot, MSD curve, repeated-seed MSD comparison.	`active_crystal_like.mp4`
Active glass-like	Parameter trace, spatial snapshot, MSD curve, repeated-seed MSD comparison.	`active_glass.mp4`
Liquid-like	Parameter trace, spatial snapshot, MSD curve, repeated-seed MSD comparison.	`liquid_like.mp4`
Gas-like	Parameter trace, spatial snapshot, MSD curve, repeated-seed MSD comparison.	`gas_like.mp4`

How the plots should be read

The order parameter and neighbor-change plot describes structural organization and bond rearrangement. The snapshot shows whether the structure is clustered, connected, or dispersed at a representative time. The MSD plot measures how far particles move on average as the lag time increases. The video link is included so that the plotted interpretation can be checked against the actual time evolution.

7.2 Active Crystal Regime

The active crystal regime is obtained when strong magnetic binding is combined with enough activity to remove defects. The parameter plot in Figure 7.1 shows the clearest structural ordering among the four phases. The local topological order rises quickly from the threshold region to values near 0.9, and after the initial transient it remains high. The nearest-neighbor change rate is initially large because the Magbots are still searching for compatible local arrangements, but it then falls mostly below the practical threshold. This combination is the signature of an actively annealed ordered state: mobility is present early, but the final neighborhood structure becomes stable.

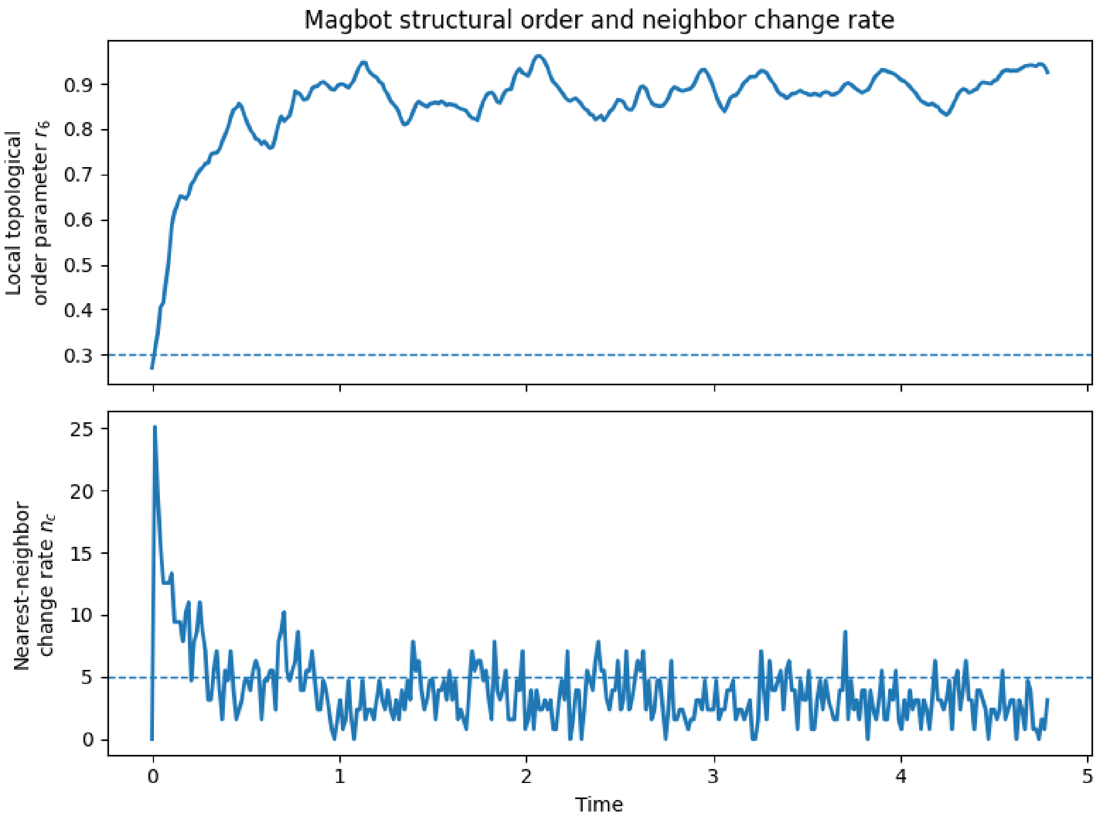

Figure 7.1: Project-generated active-crystal parameter plot. The local topological order rises rapidly and remains high, while the nearest-neighbor change rate decreases after the initial rearrangement stage.

The snapshot in Figure 7.2 confirms that this high order is not merely a numerical artifact. The Magbots form extended locally connected structures, and many units participate in larger clusters rather than remaining isolated. The moving light spot does not simply disperse the system. Instead, it helps generate local rearrangement while magnetic interactions preserve contacts. The important physical point is that activity can improve order when binding is strong enough to retain local structure. In this regime, activity plays an annealing role.

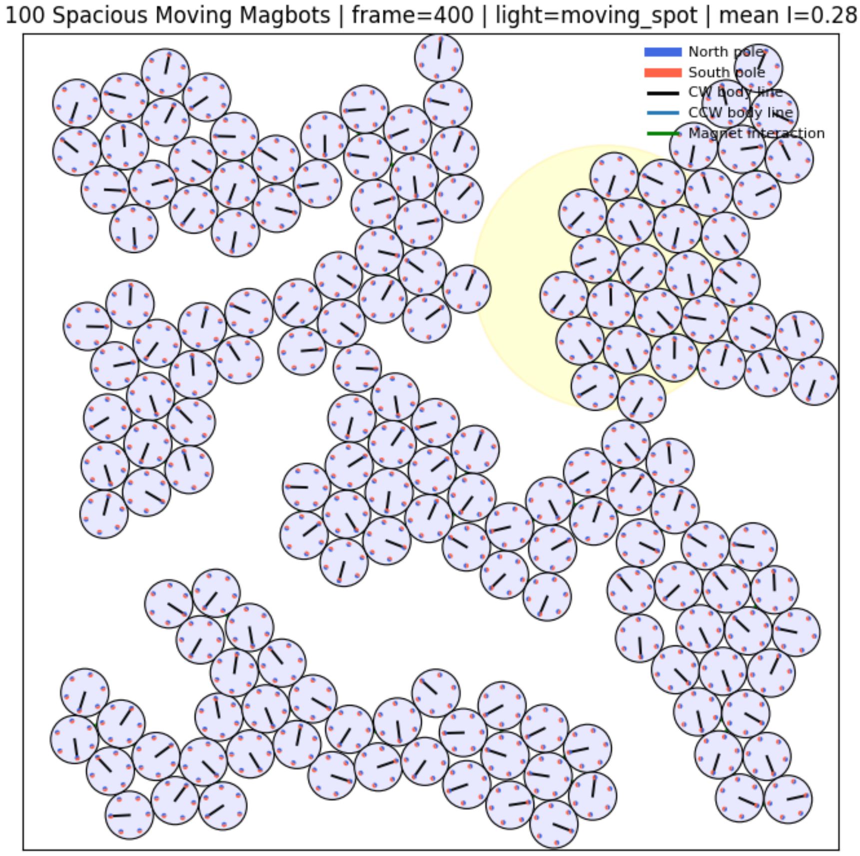

Figure 7.2: Project-generated active-crystal snapshot. The Magbots form connected locally ordered assemblies, showing that the high order parameter corresponds to visible collective organization.

The active-crystal MSD in Figure 7.3 is much smaller than the liquid-like MSD. The curve increases with oscillatory features rather than showing smooth long-range spreading. This is consistent with particles moving within a constrained ordered network. The repeated-seed analysis in Figure 7.13 gives a final MSD mean of about 3.59 at the maximum lag used for this phase, with a standard deviation of about 0.32 across three seeds. Thus the active crystal is not immobile, but its displacement is strongly bounded compared with the fluid phases.

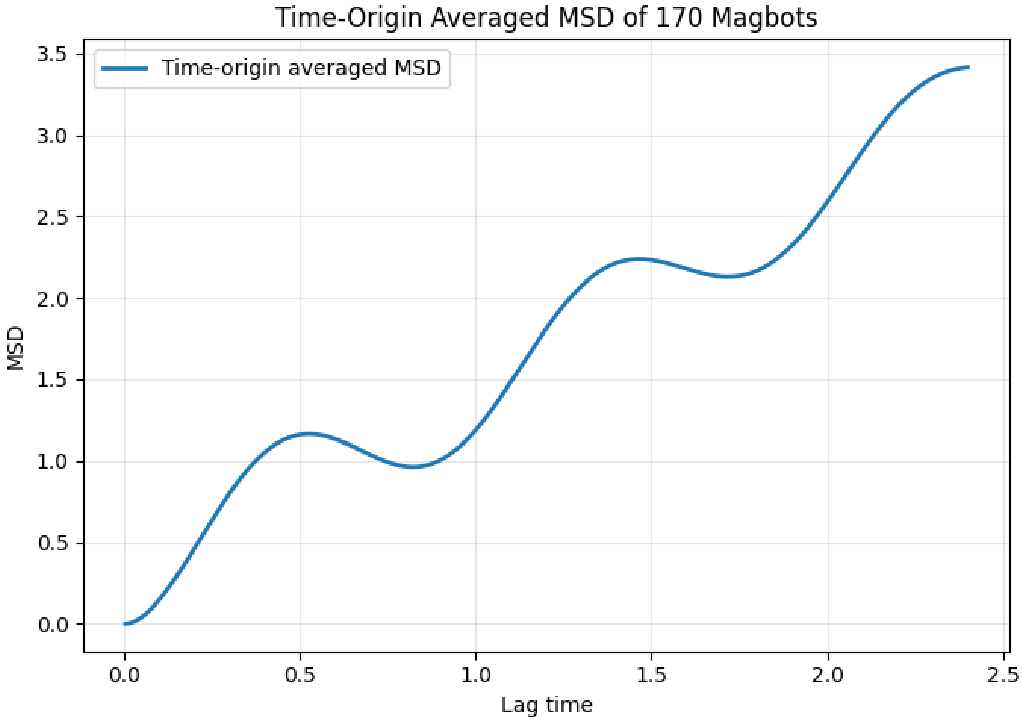

Figure 7.3: Project-generated active-crystal MSD plot. The bounded MSD growth indicates constrained motion inside an ordered network rather than free spreading.

Interpretation of the active crystal

The active crystal is the best example of constructive activity in this project. Activity is strong enough to let defects relax, while magnetic interaction is strong enough to stabilize local order. The result is not a passive crystal, but an active ordered material that continues to move internally while preserving its collective structure.

7.3 Active Glass-Like Regime

The active glass-like regime is produced when binding is strong but activity is too weak to allow complete structural relaxation. Figure 7.4 shows that the order parameter rises gradually and eventually reaches a high value, but the neighbor-change rate rapidly collapses toward zero. The system therefore develops local structure but loses the ability to reorganize. This is the central difference between the glass-like and crystal-like regimes. Both can have local order, but the glass-like phase is dynamically arrested.

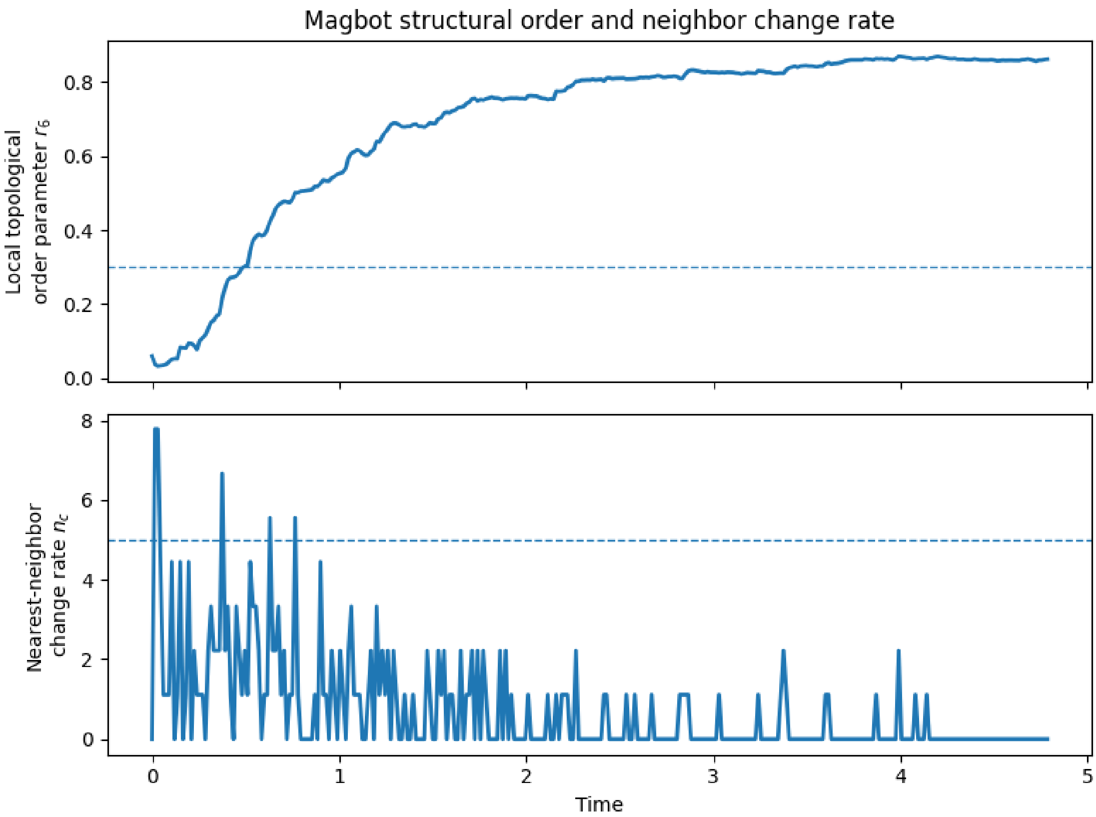

Figure 7.4: Project-generated active glass-like parameter plot. Local order increases, but the nearest-neighbor change rate decays toward zero, indicating kinetic arrest.

The snapshot in Figure 7.5 shows several compact clusters separated by empty regions. The clusters are not random isolated particles, so the state is not gas-like. However, the structure also does not form one clean extended crystal across the domain. The picture is therefore consistent with local binding and partial organization, but with insufficient rearrangement to merge or anneal all domains into a more ordered state.

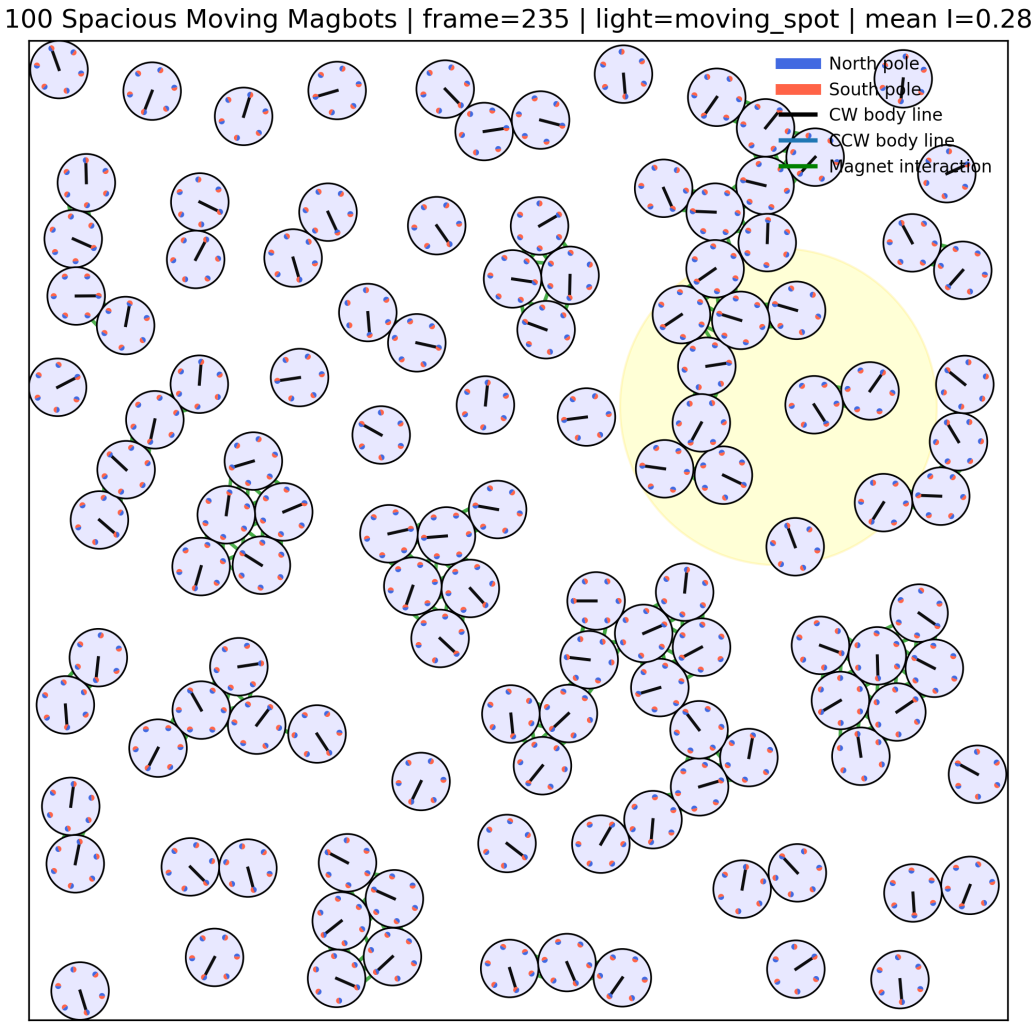

Figure 7.5: Project-generated active glass-like snapshot. The system contains locally organized clusters, but the arrangement remains domain-like and kinetically constrained.

The MSD plot in Figure 7.6 increases more smoothly than the active-crystal MSD, but the displacement remains small on the same short-lag scale. The repeated-seed summary gives a final MSD mean of about 2.62 with standard deviation about 0.19 for this phase. The low variation across seeds suggests that the constrained mobility is a robust outcome of the selected glass-like parameters. The physical interpretation is that particles can move locally, but strong binding and low activation prevent large-scale restructuring.

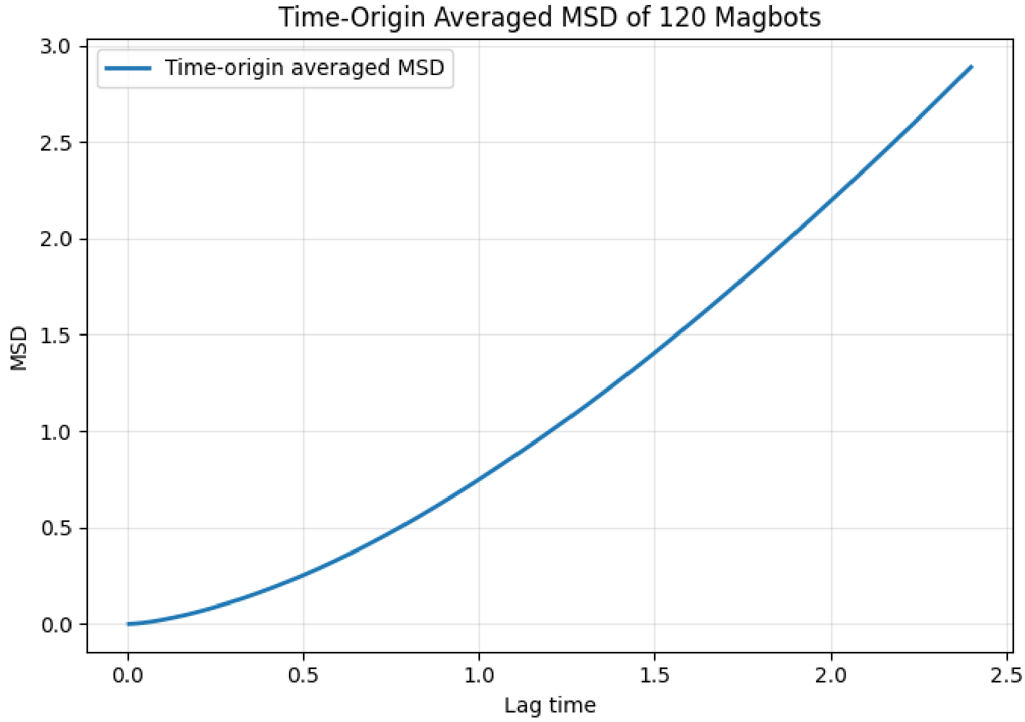

Figure 7.6: Project-generated active glass-like MSD plot. The slow growth of MSD is consistent with restricted motion inside a kinetically arrested clustered state.

Interpretation of the active glass-like state

The glass-like state proves why order alone is not enough for phase classification. The system can look locally organized while still being dynamically poor. Its defining feature is not the absence of structure; it is the loss of rearrangement before the structure can anneal into a cleaner crystal.

7.4 Liquid-Like Regime

The liquid-like regime is obtained when activity and binding become comparable. In Figure 7.7, the order parameter starts low and rises gradually, crossing the practical threshold only after a long transient. At the same time, the nearest-neighbor change rate remains high and fluctuating around the threshold. This is the clearest evidence of a flowing active material: local contacts exist, but they are continuously created and destroyed.

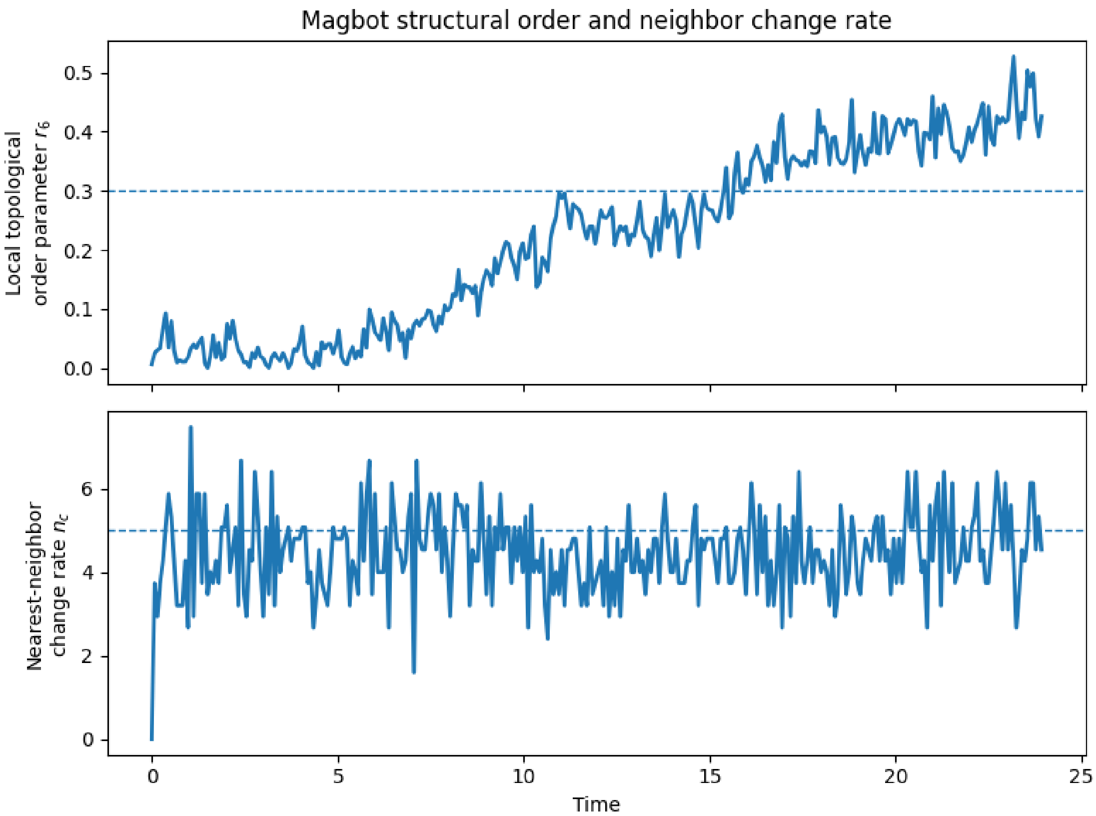

Figure 7.7: Project-generated liquid-like parameter plot. Local order grows slowly, while neighbor exchange remains persistent throughout the simulation.

The snapshot in Figure 7.8 supports the same interpretation. The Magbots are not as uniformly isolated as in the gas-like case; small pairs and short clusters are visible. However, the system does not form large stable domains like the active crystal or glass-like cases. This intermediate spatial structure is exactly what is expected for a liquid-like active material. It has short-lived local coordination but no permanent network.

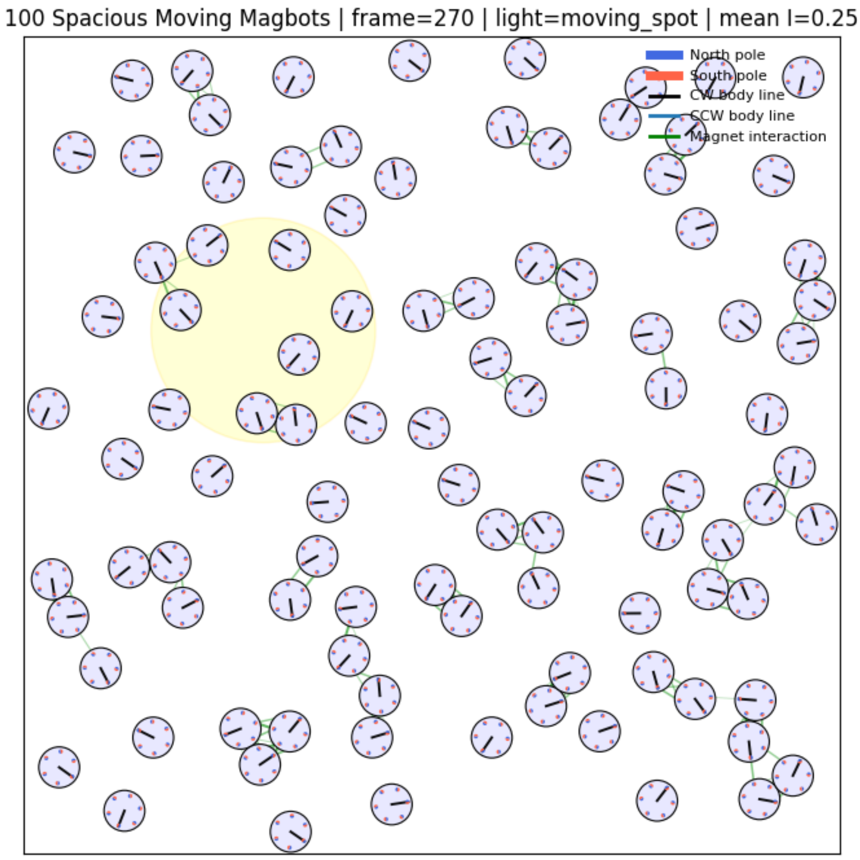

Figure 7.8: Project-generated liquid-like snapshot. The configuration contains transient pairs and small clusters rather than a stable extended assembly.

The MSD result in Figure 7.9 is the strongest mobility result in the project. The curve grows to about 66 for the representative run, much larger than the active-crystal or active-glass values. Small oscillations appear because the motion is chiral and because repeated encounters modify the displacement growth, but the overall trend is strongly increasing. The repeated-seed study gives a final MSD mean of about 71.37 with standard deviation about 3.60, showing that high mobility is repeatable across the tested seeds. This confirms that the liquid-like phase is not just visually disordered; it is dynamically fluid.

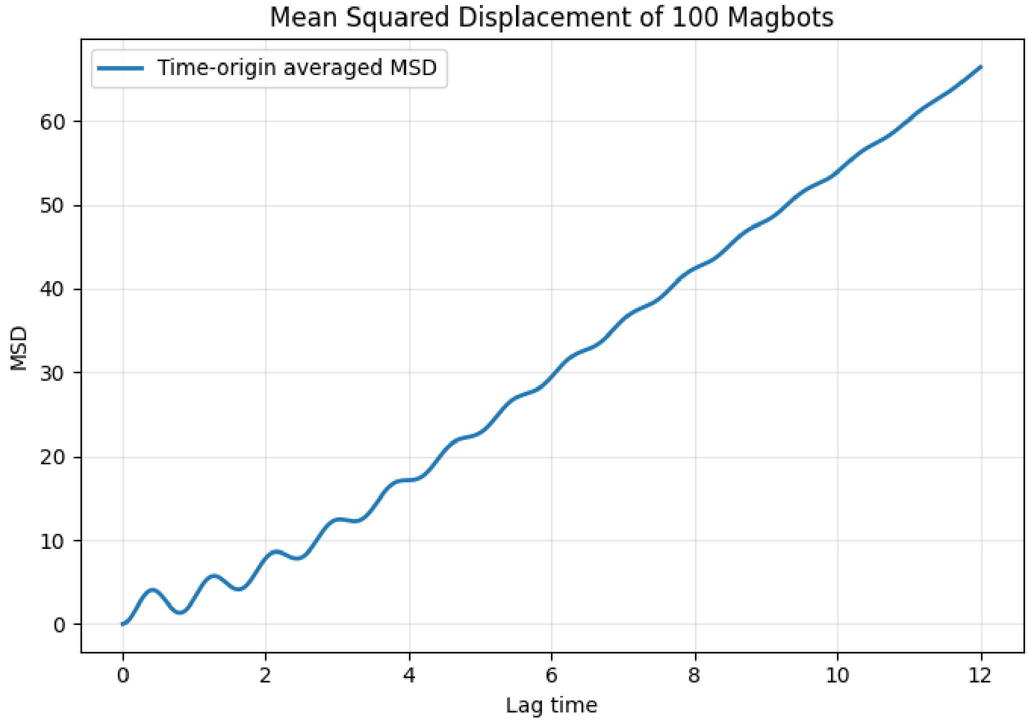

Figure 7.9: Project-generated liquid-like MSD plot. The large MSD growth demonstrates sustained displacement and fluid-like rearrangement.

Interpretation of the liquid-like state

The liquid-like regime is the most useful state for adaptive reconfiguration. It preserves enough interaction to create temporary structure, but it remains mobile enough to flow, search, and reorganize. For Robo-Matter applications, this regime is important because it lies between rigid order and complete dispersion.

7.5 Gas-Like Regime

The gas-like regime appears when activity dominates over magnetic binding. Figure 7.10 shows that the local topological order remains far below the threshold. The nearest-neighbor change rate remains active and fluctuating because particles continue to collide and separate, but these encounters do not produce stable structural order. This distinguishes the gas-like phase from the liquid-like phase: both are mobile, but only the liquid-like phase develops meaningful local coordination.

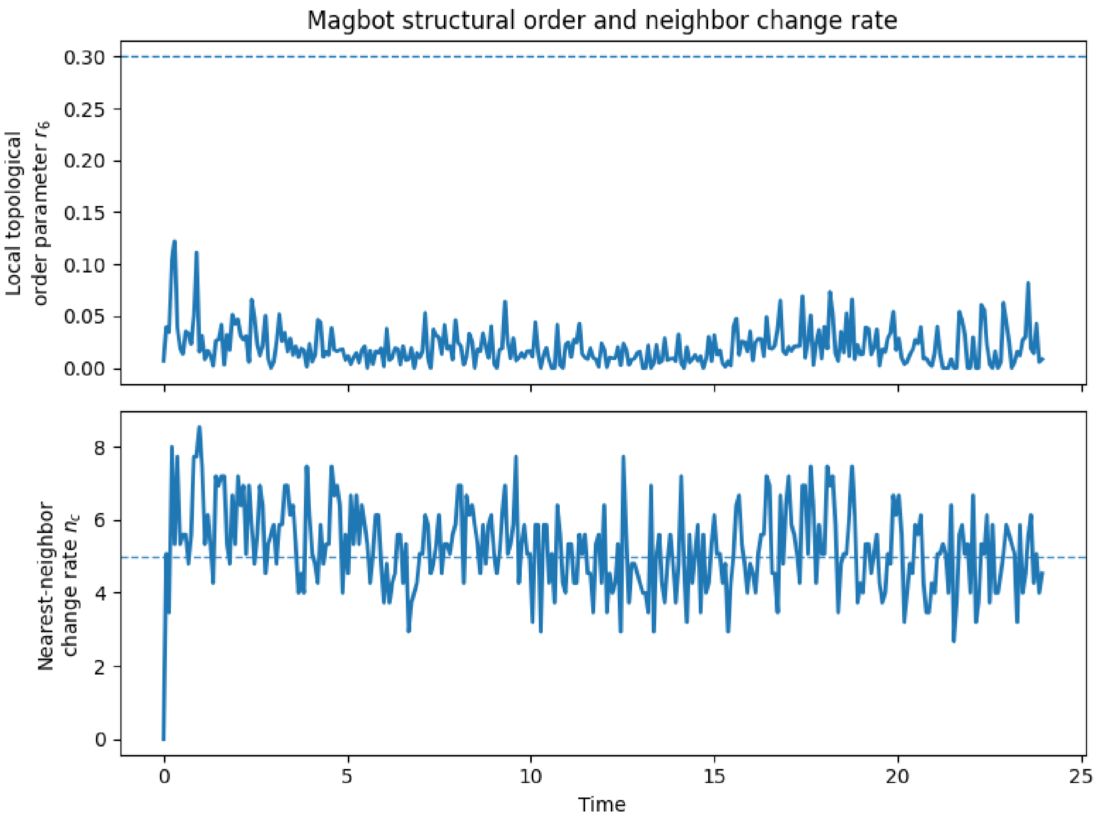

Figure 7.10: Project-generated gas-like parameter plot. The order parameter remains very low while neighbor exchange stays active, indicating mobile disorder.

The snapshot in Figure 7.11 shows a widely dispersed configuration. Most Magbots are isolated or only briefly close to another unit. The moving light region changes local activation, but weak magnetic binding is insufficient to create stable clusters. This visual evidence agrees with the low order parameter and confirms that the gas-like state is structurally sparse.

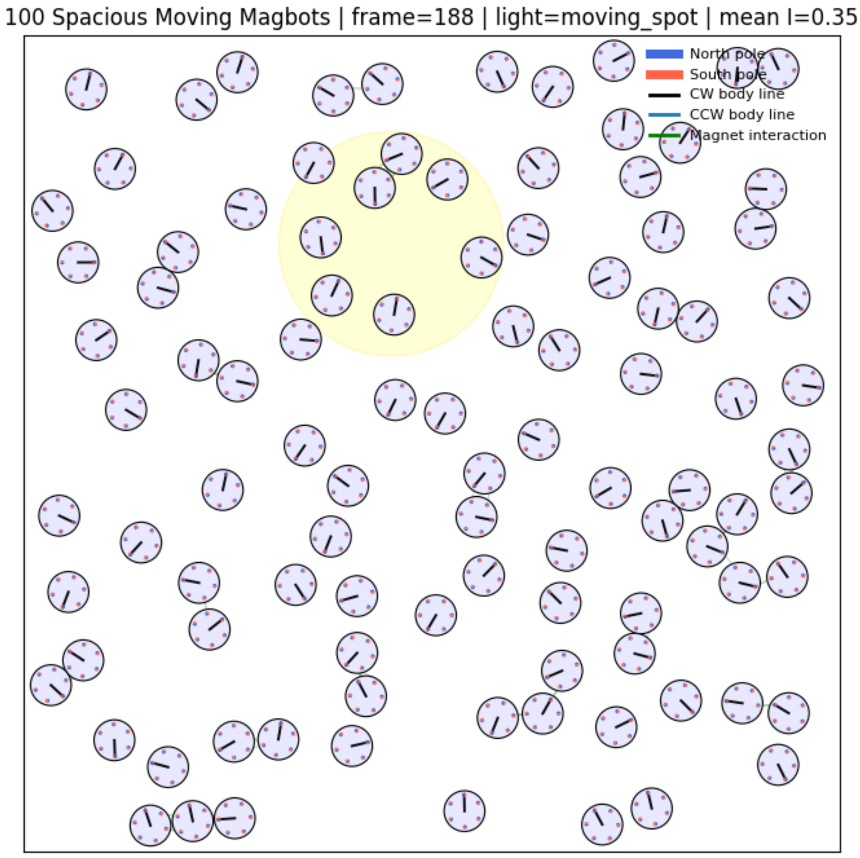

Figure 7.11: Project-generated gas-like snapshot. The Magbots remain widely dispersed, with little evidence of stable collective organization.

The gas-like MSD in Figure 7.12 grows substantially, reaching about 41 in the representative run. Its repeated-seed mean is about 27.21, but the standard deviation is large, about 11.59. This larger variability is physically reasonable because the gas-like regime is more sensitive to boundary encounters, initial positions, and chance collisions. The state is mobile, but its motion is less collectively regulated than the liquid-like state. Thus gas-like behavior should not be interpreted as a useful flowing material; it is better described as active dispersion.

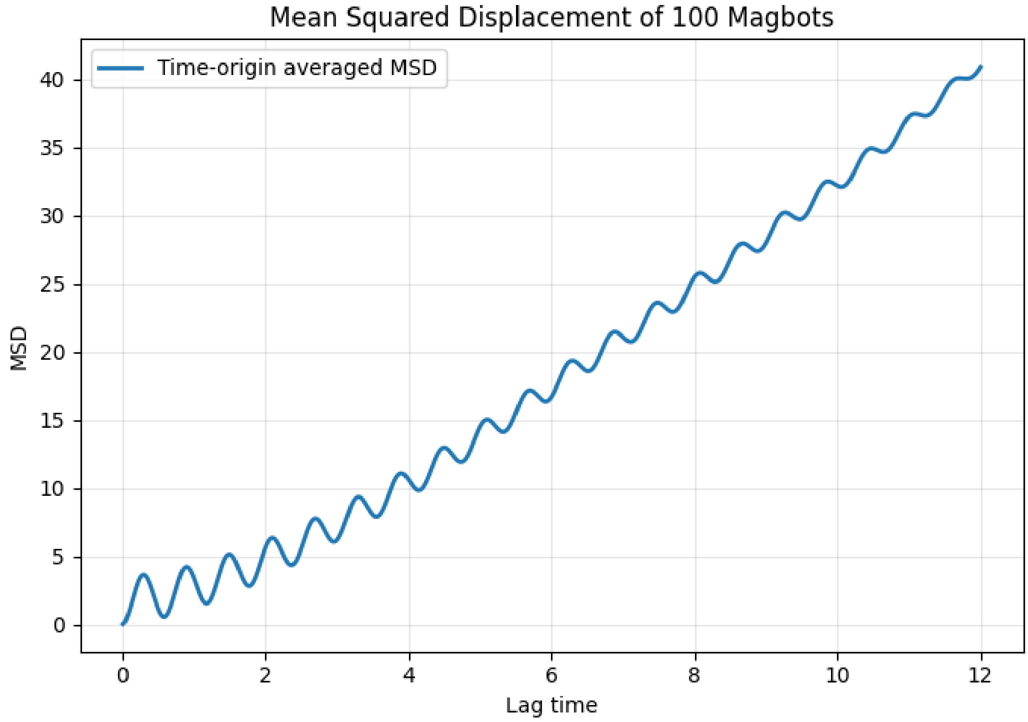

Figure 7.12: Project-generated gas-like MSD plot. The MSD grows because particles remain mobile, but the parameter plot and snapshot show that this mobility is not accompanied by stable ordering.

Interpretation of the gas-like state

The gas-like phase is an activity-dominated state. It is mobile, but it lacks useful cohesion. In a Robo-Matter context, this phase may help spread agents through space, but by itself it does not provide stable material functionality.

7.6 Repeated-Seed MSD and Cross-Phase Comparison

To address repeatability, each phase was simulated for three random seeds using the MSD scripts. The resulting mean and standard-deviation envelopes are shown in Figure 7.13. This repeated-seed analysis does not replace a full phase diagram, but it gives an important check that the qualitative mobility ranking is not based on one accidental trajectory. The liquid-like phase consistently shows the largest displacement growth, the active crystal and active glass-like phases remain constrained, and the gas-like phase shows mobile but more variable behavior.

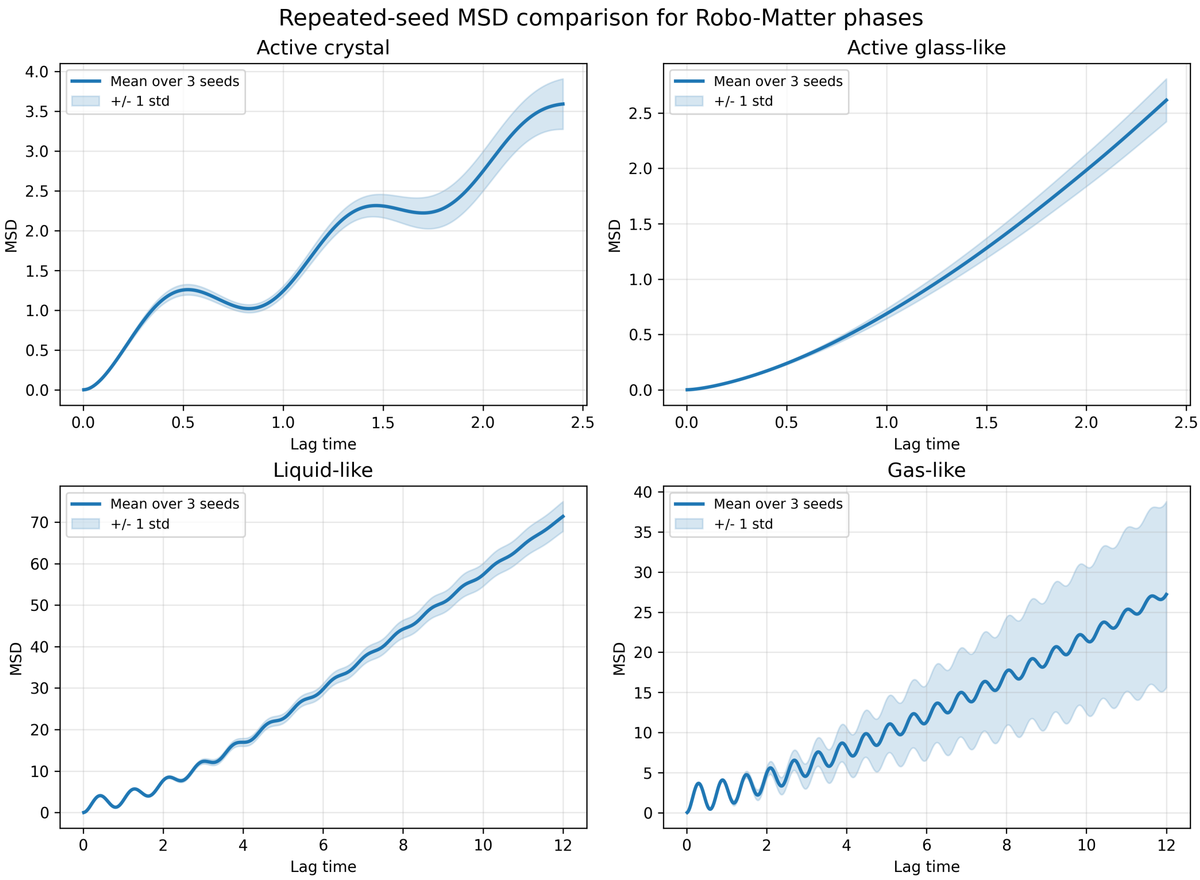

Figure 7.13: Project-generated repeated-seed MSD comparison. Each phase was repeated for seeds 7, 17, and 29. The curve shows the mean MSD and the shaded region shows one standard deviation.

Table 7.2: Final repeated-seed MSD values at the maximum lag used for each phase.

Regime	Seeds	Maximum lag	Final MSD mean	Interpretation
Active crystal	3	2.40	3.59	Ordered and mobile only within a constrained network.
Active glass-like	3	2.40	2.62	Most dynamically restricted among the tested phase classes.
Liquid-like	3	12.00	71.37	Strong repeated mobility with persistent rearrangement.
Gas-like	3	12.00	27.21	Mobile but highly seed-dependent and structurally disordered.

The repeated-seed MSD comparison separates two ideas that can otherwise be confused. The gas-like state is disordered and mobile, but its final MSD varies strongly across seeds. The liquid-like state is also mobile, but its mobility is more consistently high in the repeated runs. This difference matters for robotic material design. If the goal is controlled reconfiguration, a liquid-like state is more useful than a gas-like state because it combines mobility with interaction. If the goal is rapid dispersal, the gas-like state may be useful, but it sacrifices cohesion.

The active crystal and glass-like curves are plotted over a shorter maximum lag because their simulations use a smaller time step. Both remain in a low-MSD range compared with the liquid and gas cases. The difference between them is therefore not mainly the magnitude of MSD. It lies in the relation between structure and rearrangement. The active crystal reaches high order after an early rearrangement stage, while the glass-like system becomes locally ordered but dynamically arrested.

Table 7.3: Metric-based classification of collective states.

Metric combination	Physical meaning	Likely regime
High order, low n_c	Stable local structure with little rearrangement.	Active crystal or arrested glass
High/medium order, high n_c	Structured but flowing network.	Liquid-like
Low order, high/medium n_c	Disordered mobile particles or small clusters.	Gas-like
Low order, low n_c	Frozen disordered state or insufficient motion.	Deep glass-like limit

The evidence from Figures 7.1–7.12 can be summarized as a competition between binding and agitation. Strong binding with enough activity produces an active crystal. Strong binding with too little rearrangement produces a glass-like arrested state. Comparable binding and activity produces a flowing liquid-like state. Weak binding under high activity produces an active gas.

$$\begin{aligned} &\text{glass-like} \xrightarrow{\text{increase activity}} \text{active crystal} \xrightarrow{\text{increase activity further}} \text{liquid-like} \\ &\qquad\qquad \xrightarrow{\text{weak binding or high activity}} \text{gas-like.} \end{aligned} \tag{7.1}$$

The first transition is subtle and important. A small increase in activity can improve order by allowing particles to escape trapped local configurations. However, too much activity breaks bonds faster than they can stabilize, producing liquid-like or gas-like behavior. This is why activity should not be described only as "noise" or "temperature." In active Robo-Matter, activity is also a mechanism of search, defect correction, and controlled reconfiguration.

7.7 Model Assessment

The model is useful because it is physically interpretable: every term in the update rule corresponds to activity, force, torque, noise, or environmental control. It includes both translational and rotational dynamics, treats the internal magnet phases as active variables rather than fixed labels, reproduces four distinct collective regimes under parameter changes, and remains extendable because local neighbor search keeps the computation manageable.

The model is also deliberately simplified. The magnetic potential is not a full dipole-field calculation, friction and vibration-motor dynamics are represented phenomenologically, and the light-field drift is a compact control approximation rather than a detailed photo-electromechanical model. The repeated-seed MSD analysis improves the reliability of the mobility interpretation, but stronger quantitative phase classification would still require longer simulations, more random seeds, additional structural metrics, and direct comparison with experimental datasets.

7.8 Discussion

The main scientific outcome of the project is the demonstration that Robo-Matter-like behavior can be generated from simple local rules and then classified through multiple diagnostics. The earlier version of the result could be read as four example plots. The expanded analysis shows a more complete story: the same model produces visibly different structures, different order-parameter histories, different neighbor-exchange behavior, different MSD growth, and different repeated-seed mobility trends.

The most important physical conclusion is that the useful Robo-Matter regime is not located at the extremes. If magnetic binding dominates too strongly, the system becomes arrested. If activity dominates too strongly, the system disperses. Between these limits, activity can be constructive. It can help Magbots rearrange, repair defects, and explore configurations while magnetic interactions preserve local cohesion. This balance is what makes active robotic matter different from both ordinary passive materials and ordinary robot swarms.

The simulation videos are also scientifically useful because they show the temporal path to each final state. The active crystal video shows clustering and rearrangement before stabilization. The active glass video shows local cluster formation followed by reduced rearrangement. The liquid-like video shows continual bond exchange, and the gas-like video shows dispersed motion. These video observations agree with the quantitative plots and strengthen the phase interpretation.

8 Conclusion

8.1 Summary, Limitations, and Future Directions

This project developed a two-dimensional simulation framework for Robo-Matter inspired by active magnetic Magbot systems. Each Magbot was represented as a rigid circular disk with six magnetic sites, chiral rotation, active translational motion, light-field modulation, magnetic force, internal magnet torque, body torque, noise, steric repulsion, and reflecting boundaries. The simulation was implemented in Python and used cell-list neighbor search for efficient local interaction evaluation.

The main contribution of the thesis is not only that four representative Robo-Matter states were generated, but that those states were interpreted using several mutually supporting diagnostics. The final results chapter includes structural order and neighbor-change plots, spatial snapshots, MSD curves, repeated-seed MSD comparisons, and links to phase videos. This gives a stronger basis for classification than a small number of final figures. The active crystal is identified by high local order and reduced neighbor exchange after an initial rearrangement stage. The active glass-like state is identified by increasing local order combined with a collapse of neighbor exchange, indicating kinetic arrest. The liquid-like state is identified by persistent neighbor exchange and the largest repeated-seed MSD growth. The gas-like state is identified by very low order, dispersed snapshots, and mobile but seed-sensitive MSD growth.

The results show that activity has a dual role. At moderate levels it improves ordering by allowing particles to escape local traps and anneal defects. At high levels, or when binding is too weak, it destroys sustained contacts and produces gas-like dispersion. Magnetic interaction also has a dual role. It is necessary for material-like cohesion, but if the system cannot rearrange, strong binding can freeze the structure into a glass-like state. The useful adaptive regime therefore lies between rigid arrest and complete dispersion, where the system keeps enough cohesion to act like a material while retaining enough mobility to reconfigure.

The repeated-seed MSD analysis strengthens the mobility interpretation. For three seeds, the liquid-like phase gives the largest final MSD mean, about 71.37 at the maximum lag, while the active crystal and active glass-like phases remain below 4 on their shorter lag interval. The gas-like state remains mobile but has a much larger seed-to-seed variation, which reflects its sensitivity to initial conditions, boundary encounters, and transient collisions. This comparison supports the conclusion that liquid-like Robo-Matter is a more controlled reconfigurable state than gas-like dispersal.

The limitations of the model should be kept clear. The magnetic interaction is simplified, the light response is phenomenological, the contact mechanics are not a full hardware model, and the current repeated-seed study is still small. The plots should therefore be interpreted as qualitative and mechanism-level evidence. Future work should extend the repeated-seed analysis to larger ensembles, perform systematic parameter sweeps, compute bond-orientational order, radial distribution functions, cluster-size distributions, and long-time MSD exponents, and compare the simulated phase boundaries more directly with experimental Robo-Matter measurements.

Overall, the project shows that Robo-Matter behavior can emerge from local active magnetic rules without prescribing global order. The thesis demonstrates how ordered, arrested, flowing, and dispersed states arise from the same model by changing the balance between activity and binding. This provides a useful computational bridge between active matter physics and reconfigurable robotic materials, and it gives a clear foundation for future studies on feedback control, adaptive morphing, and programmable material response.

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[4] S. Kumar, "Suraj BTP project: Robo-Matter duality," GitHub repository, 2026. Available: `https://github.com/suraj4556kumar-sketch/Robo_matter_duality/tree/main/BTP-2`.

A Simulation Parameters

A.1 Default Magbot Geometry

Table A.1: Core geometric parameters used in the simulation.

Parameter	Value	Meaning
N_{bots}	100–150	Number of simulated Magbots
N_{mag}	6	Magnetic sites per Magbot
R_{bot}	1.0	Radius of each circular body
R_{ring}	0.78	Radius of the magnetic-site ring
r_{mag}	0.105	Visual/display radius of magnetic sites

A.2 Common Numerical Parameters

Table A.2: Common time integration and noise parameters.

Parameter	Typical value	Meaning
D_t	0.012	Translational noise strength
D_r	0.025	Rotational noise strength
D_ϕ	0.012	Internal magnet phase noise
μ_t	1.0	Translational mobility
μ_r	0.50–0.55	Rotational mobility
μ_ϕ	1.0	Magnet phase mobility
Δt	0.003–0.015	Time step
steps	900–1600	Number of frames/updates
seed	7	Random seed used in representative runs

B Mathematical Derivations

B.1 Euler-Maruyama Discretization

For a stochastic differential equation

$$dX = a(X,t)\,dt + b\,dW_t, \tag{B.1}$$

the Euler-Maruyama update is

$$X_{n+1} = X_n + a(X_n,t_n)\Delta t + b\sqrt{\Delta t}\,\eta_n, \tag{B.2}$$

where η_n is a standard normal random variable. This is the basis for the translational, rotational, and internal magnet phase updates used in the simulation.

B.2 Torque from Off-Center Force

For a force $\mathbf{F} = (F_x, F_y)$ applied at lever arm $\mathbf{r} = (r_x, r_y)$ from the center of a two-dimensional rigid body, the out-of-plane torque is

$$\tau_z = r_x F_y - r_y F_x. \tag{B.3}$$

This expression is implemented through the `cross2d` helper function in the project scripts.

C Algorithm and Code Structure

C.1 Main Algorithm

The simulation begins by initializing Magbot positions, body orientations, chirality values, and internal magnet phases. At every time step, the code computes the local light intensity and light gradient, builds a cell list for neighbor detection, evaluates steric repulsion and magnetic forces for nearby Magbot pairs, and computes the associated body and internal magnet torques. The positions, body angles, and magnet phases are then updated by Euler-Maruyama integration, followed by reflecting-wall handling and overlap resolution. The final part of each update refreshes visualization elements such as bodies, magnets, heading lines, light spots, and interaction lines, while the stored state is used for order and neighbor-change analysis.

C.2 Representative Update Pseudocode

Listing C.1: Simplified pseudocode representing the update logic used in the simulation.

```
for frame in range(steps):
    t = frame * dt

    I, gradI = light_intensity_and_gradient(pos, t)
    F, T_body, T_phi = compute_forces_and_torques(pos, theta, phi
        )

    n = np.column_stack((np.cos(theta), np.sin(theta)))
    v = v_base + v_light_gain * I
    omega = chirality * (omega_base + omega_light_gain * I)

    F_light = light_drift_strength * gradI

    pos += (mu_t * F + F_light + v[:, None] * n) * dt
    pos += np.sqrt(2.0 * Dt * dt) * rng.normal(size=pos.shape)

```

```
    theta += (omega + mu_r * T_body) * dt
    theta += np.sqrt(2.0 * Dr * dt) * rng.normal(size=len(theta))

    phi += mu_phi * T_phi * dt
    phi += np.sqrt(2.0 * Dphi * dt) * rng.normal(size=phi.shape)

    apply_reflecting_boundaries(pos, theta)
    resolve_overlaps(pos)
```

C.3 Source Files Used

Table C.1: Main local files used as project material.

File	Role in the report
robo_matter.pdf	Explanation of Magbot variables, forces, torque, light field, and parameter meaning.
s41467-024-53123-6...pdf	Primary research paper on experimental Robo-Matter, stored in the Research Papers folder.
states/code_phases/*.py	Phase simulation scripts for active crystal, active glass-like, liquid-like, and gas-like regimes.
states/MSD/*.py	MSD scripts used to compute time-origin averaged displacement curves and repeated-seed MSD comparisons.
parameter_MSD_snapshot/*.png	Parameter plots, spatial snapshots, MSD plots, and repeated-seed MSD summary included in the results section.
repeated_seed_msd_summary.csv	Numerical summary of repeated-seed MSD final values used in Table 7.2.
active_crystal_like.mp4	Active-crystal simulation video.
active_glass.mp4	Active glass-like simulation video.
liquid_like.mp4	Liquid-like simulation video.
gas_like.mp4	Gas-like simulation video.
BTP-2 GitHub folder	Online repository folder containing the BTP-2 report, code, figures, and supporting material.

D Phase Notes

D.1 Parameter Influence

The parameter response is consistent with the activation-versus-binding interpretation used throughout the report. Increasing k_m promotes magnetic binding and local order, while increasing r_0 makes magnetic sites influence each other over a longer distance. Increasing v_{base} raises translational agitation, and increasing ω_{base} strengthens chiral rotation, which can help rearrangement at moderate values but destabilize bonds when too large. Increasing the box size lowers density and reduces frequent clustering, while stronger noise tends to destroy order and move the system toward gas-like behavior.

D.2 Transition Summary

Final phase summary

The simulated Robo-Matter states are controlled by the balance between magnetic binding and active agitation. Strong binding with enough rearrangement produces active crystals; strong binding with too little rearrangement produces glass-like arrest; comparable binding and activity produces liquid-like flow; weak binding under high activity produces gas-like dispersion.